

V3 Pipeline — Egitim Raporu

Tarih: 2026-03-22 19:26

PIPELINE OZELLIKLERI:

- Feature Engineering: V3 (Label-encoded predictions + ayri OHE + manuel k-mer)
- Feature sayisi: 989
- Ornek sayisi: 4287
- Model turleri: LightGBM, XGBoost
- Hyperparameter: Grid Search (12 combo/model, 3-fold CV)
- Degerlendirme: 5-Fold Stratified CV

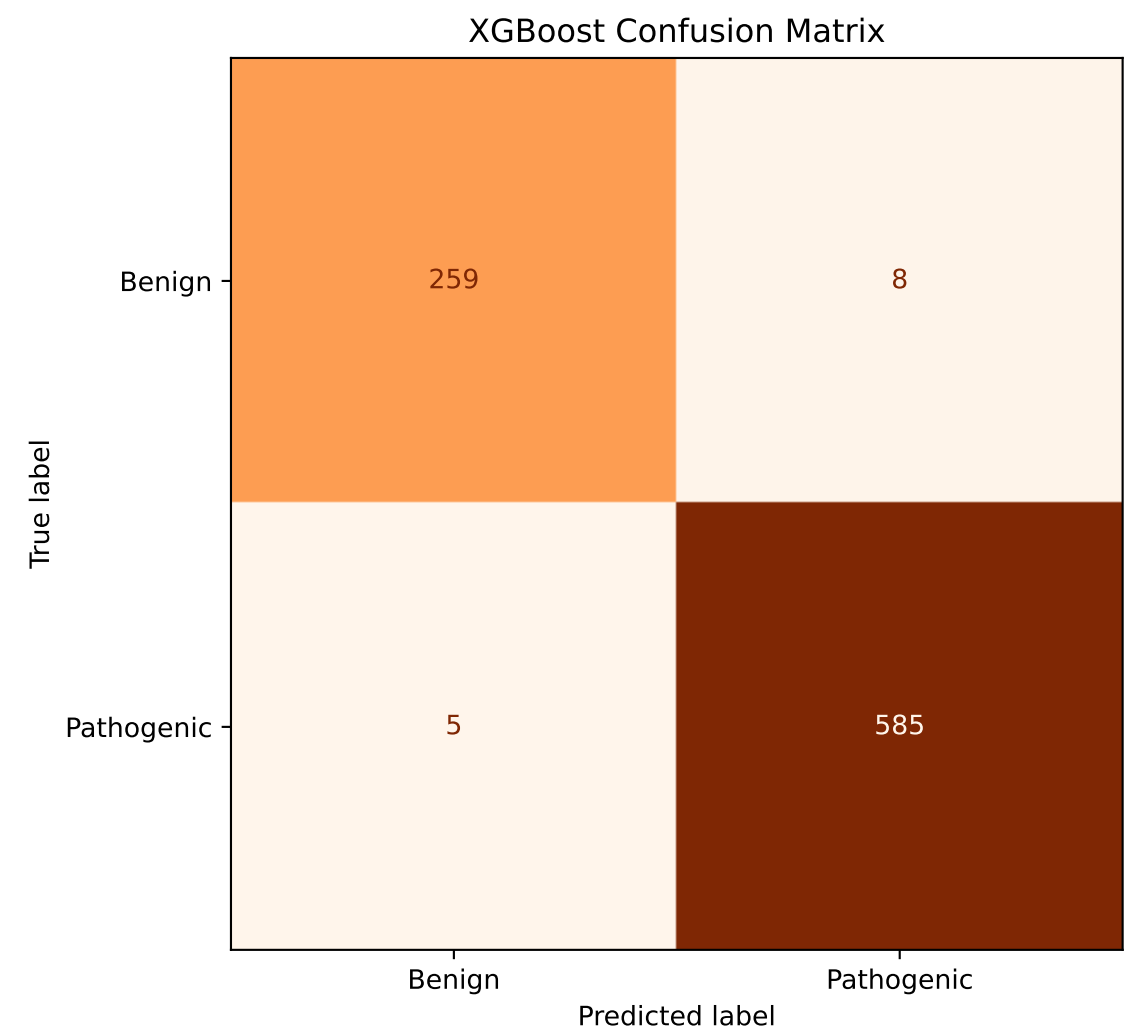
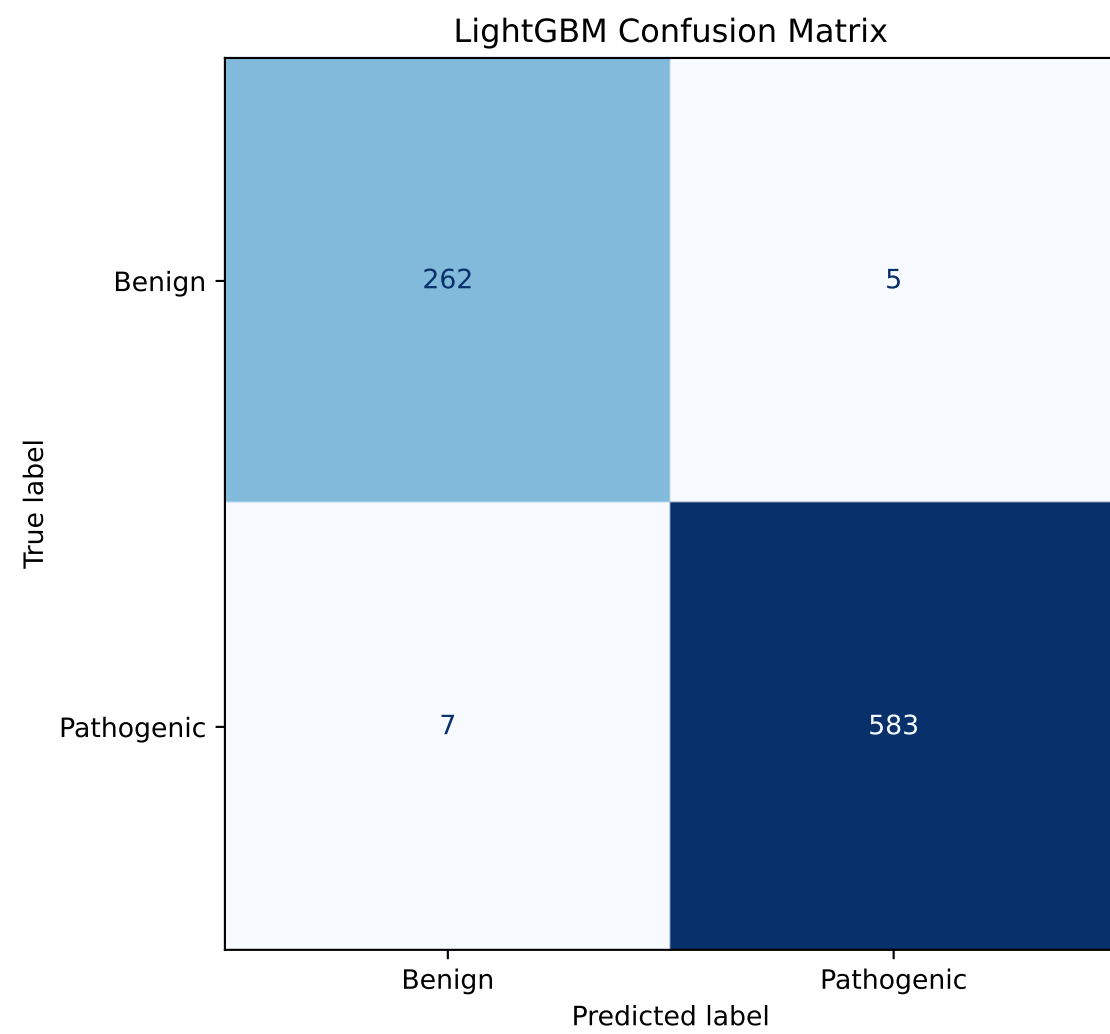
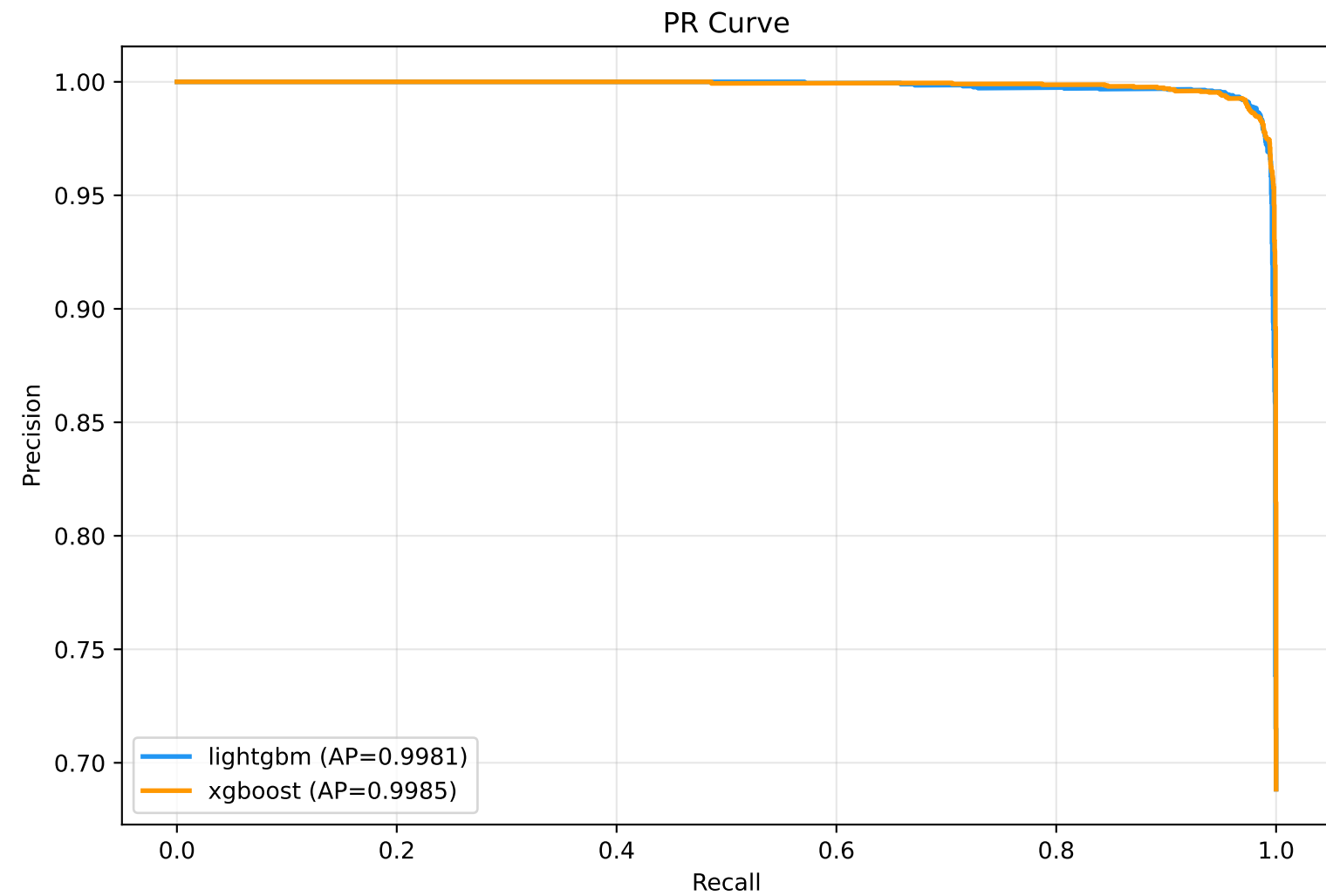
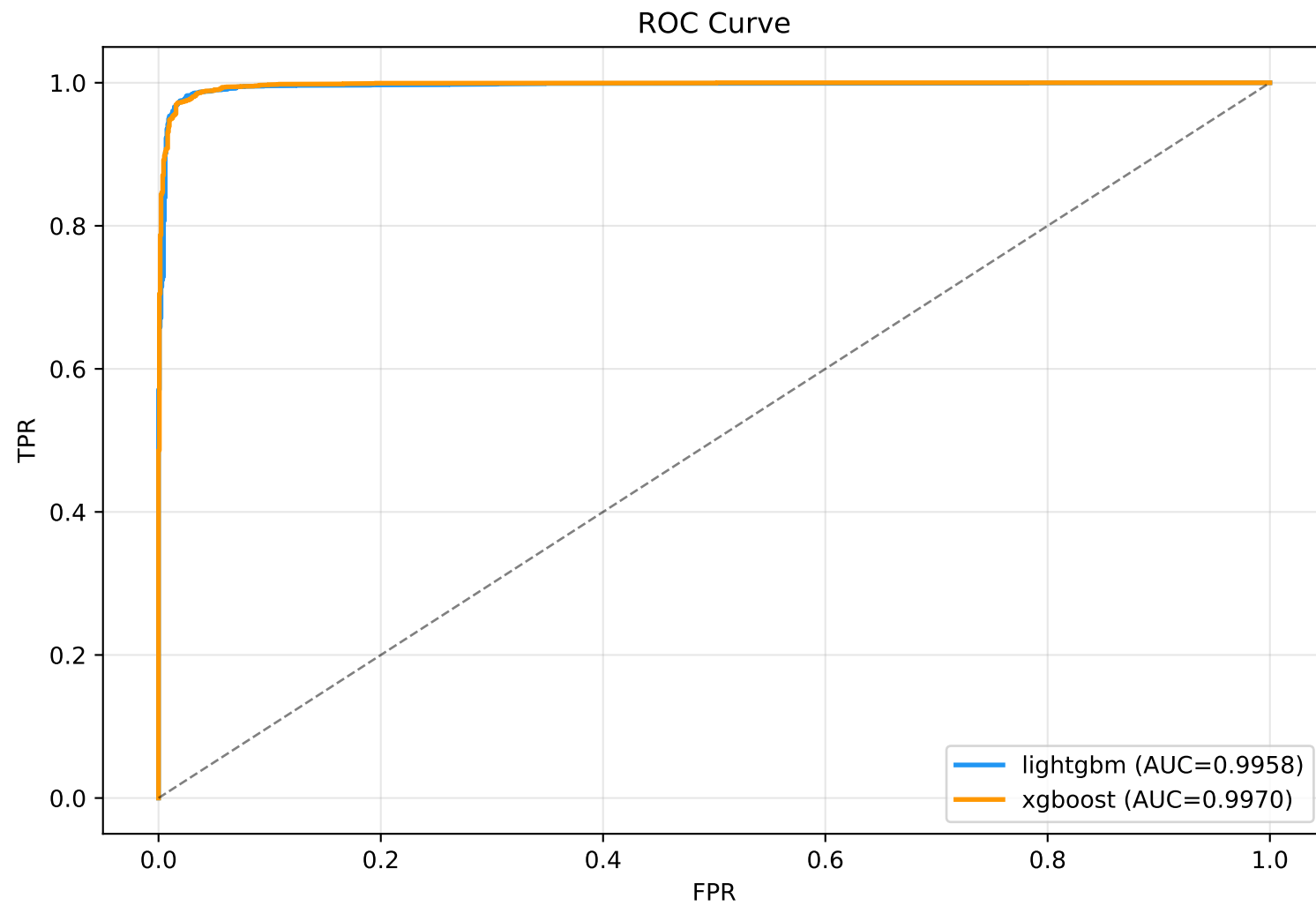
5-FOLD CV SONUCLARI:

lightgbm : F1 = 0.9863 +/- 0.0042
xgboost : F1 = 0.9863 +/- 0.0030

KARSILASTIRMA (full_fe vs v3):

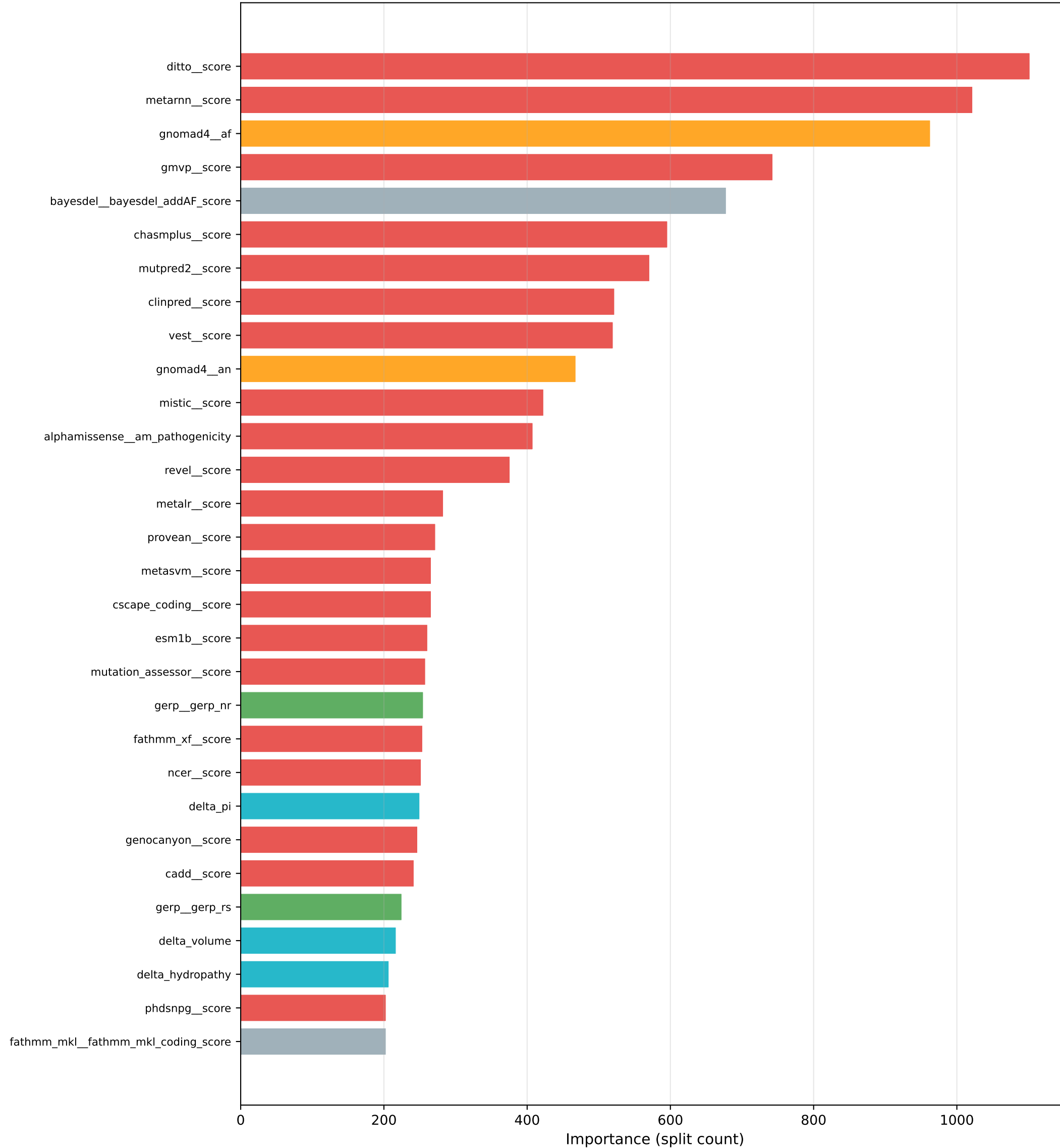
lightgbm: full_fe F1=0.6967 -> v3 F1=0.9863 (delta=+0.2896)
xgboost: full_fe F1=0.6879 -> v3 F1=0.9863 (delta=+0.2985)

V3 Pipeline — Model Performans Grafikleri

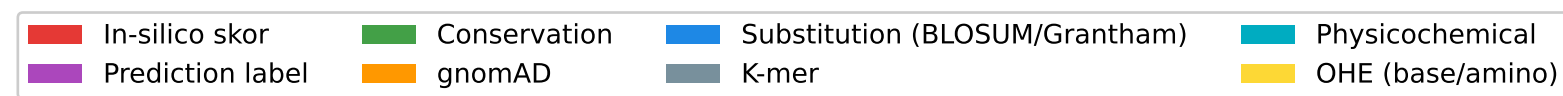
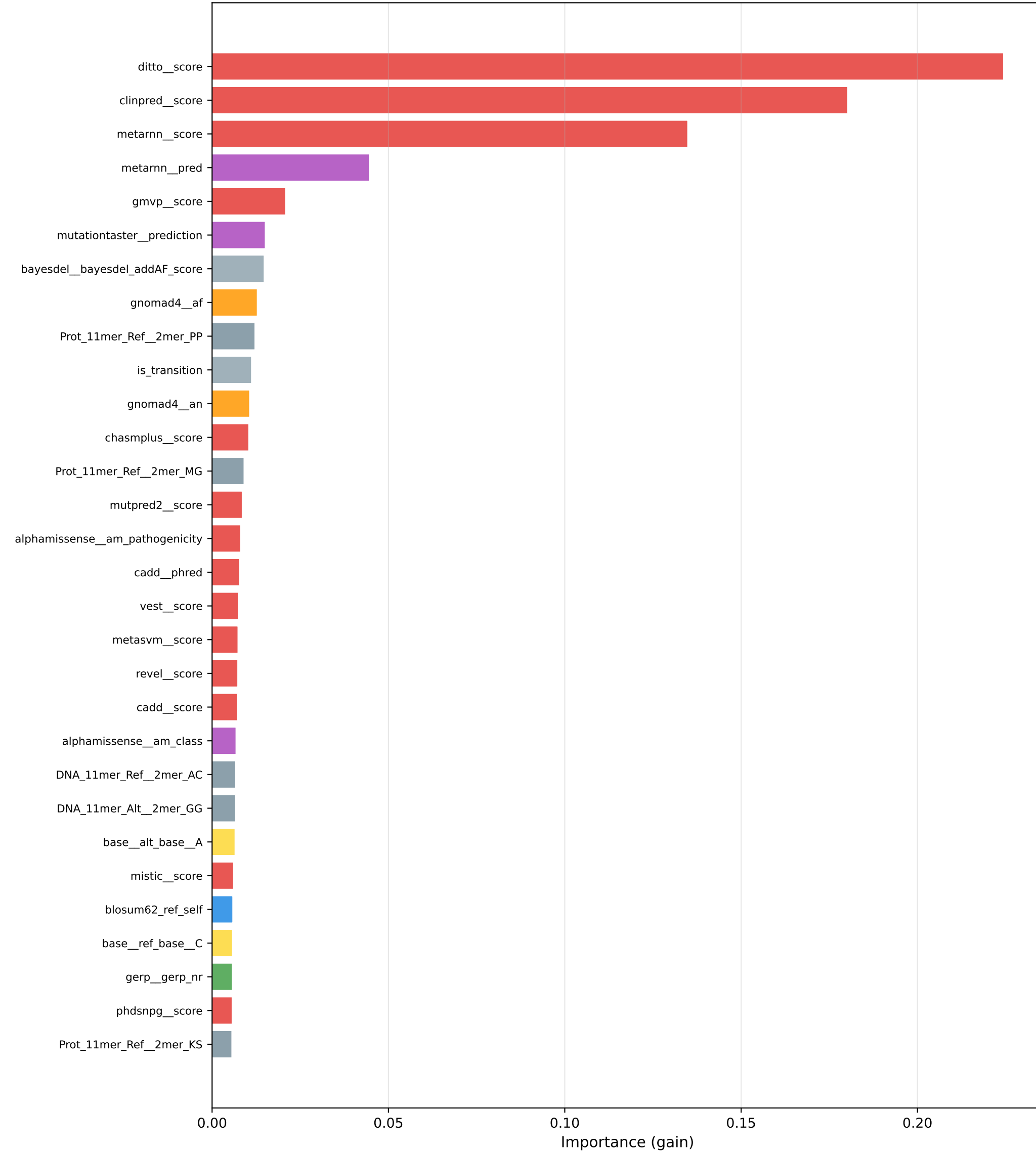


Feature Importance — Top 30 (Son Fold)

LightGBM

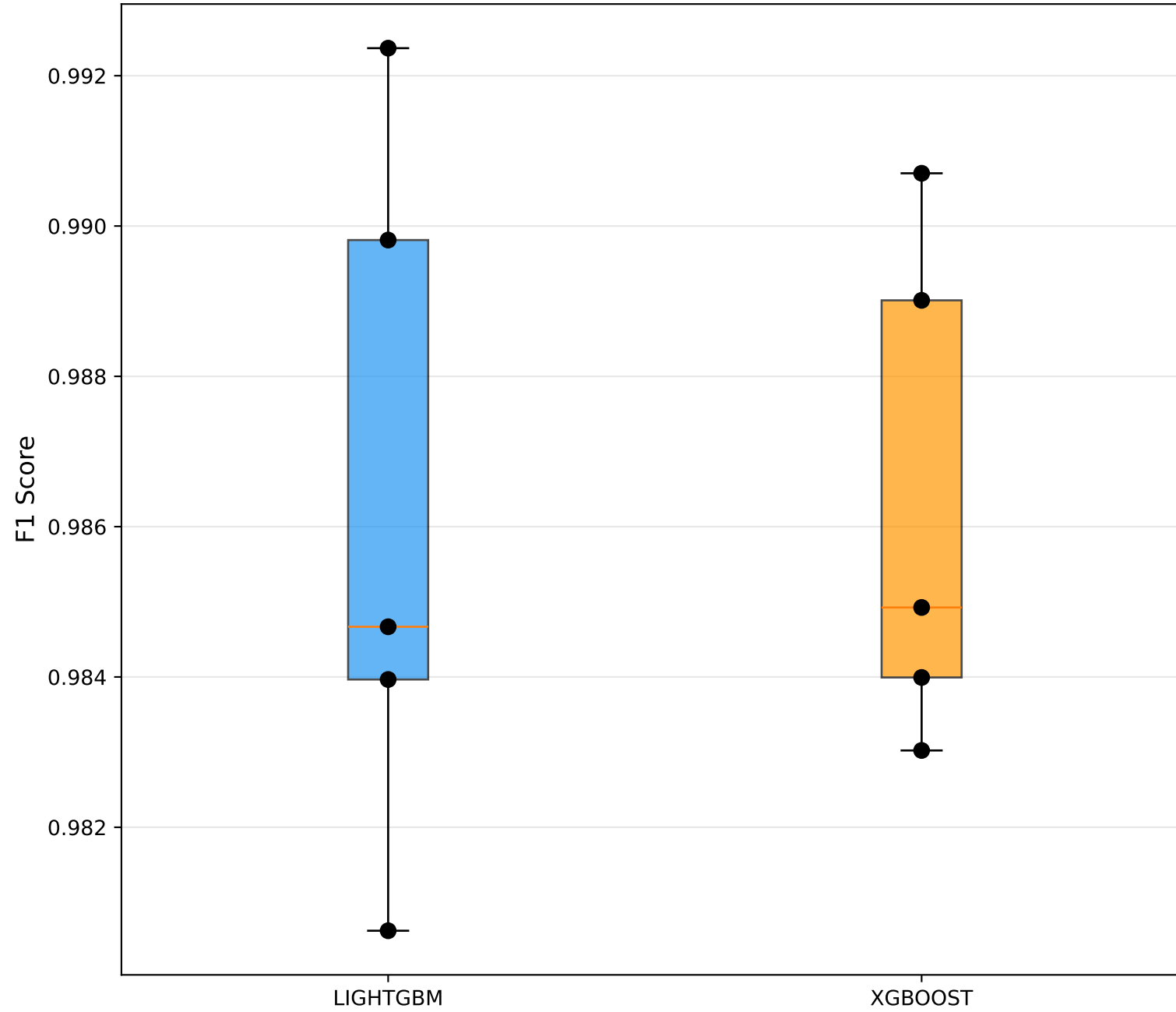


XGBoost

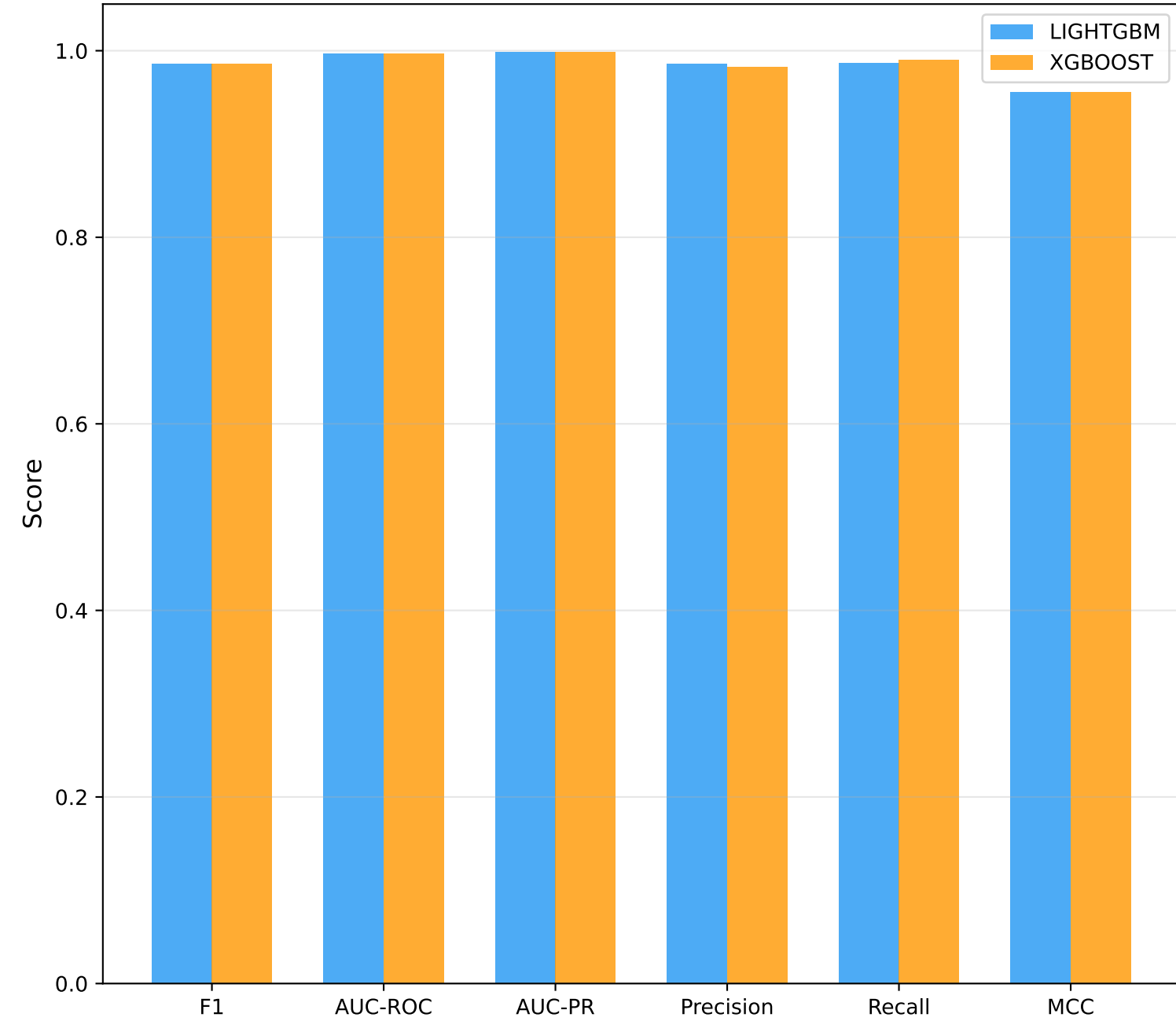


Model Karsilastirma

F1 Dagilimi (5-Fold)



Ortalama Metrikler (5-Fold)



Detayli 5-Fold CV Sonuclari

Fold	Model	F1	AUC-ROC	AUC-PR	Precision	Recall	Specificity	MCC
1	lightgbm	0.9847	0.9966	0.9986	0.9897	0.9797	0.9776	0.9516
1	xgboost	0.9830	0.9968	0.9986	0.9847	0.9814	0.9664	0.9459
2	lightgbm	0.9840	0.9961	0.9980	0.9815	0.9865	0.9588	0.9482
2	xgboost	0.9840	0.9965	0.9983	0.9799	0.9882	0.9551	0.9482
3	lightgbm	0.9806	0.9966	0.9984	0.9749	0.9864	0.9438	0.9372
3	xgboost	0.9849	0.9969	0.9986	0.9735	0.9966	0.9401	0.9510
4	lightgbm	0.9924	0.9987	0.9994	0.9932	0.9915	0.9850	0.9755
4	xgboost	0.9907	0.9988	0.9994	0.9882	0.9932	0.9738	0.9700
5	lightgbm	0.9898	0.9968	0.9987	0.9915	0.9881	0.9813	0.9674
5	xgboost	0.9890	0.9972	0.9987	0.9865	0.9915	0.9700	0.9646
ORT	lightgbm	0.9863	0.9970	0.9986	0.9862	0.9864	0.9693	0.9560
ORT	xgboost	0.9863	0.9972	0.9987	0.9826	0.9902	0.9611	0.9559